Toy Store Sales Analytics Dashboard

1. Project Overview

This project is an end-to-end Business Intelligence solution built using SQL and Power BI to analyze an e-commerce toy store dataset. The objective was to convert raw transactional data into meaningful business insights through interactive dashboards.

Business Scenario

The management team needed a centralized dashboard to monitor revenue, orders, product performance, website traffic, marketing effectiveness, conversion rate, and refund trends.

2. Business Problem Statement

The business data was distributed across multiple tables, making analysis slow and manual. A unified reporting solution was required to support faster, data-driven decision making.

3. Business Objectives

- Build an interactive Power BI dashboard.
- Monitor Revenue, Orders, Sessions, Conversion Rate, Average Order Value and Refund Amount.
- Analyze monthly sales trends.
- Evaluate product performance.
- Compare marketing channels.
- Support strategic business decisions.

4. Dataset Overview

Table	Description
Orders	Order level sales information
Order Items	Product details for each order
Products	Product master data
Website Sessions	Visitor sessions
Website Pageviews	Browsing activity
Order Item Refunds	Refund transactions

5. Data Relationships

• Orders ↔ Order Items (Order ID)
• Orders ↔ Website Sessions (Website Session ID)
• Products ↔ Order Items (Product ID)
• Orders ↔ Order Item Refunds (Order ID)

6. SQL Analysis Summary

Module	**Objective**	**Business Outcome**
Data Exploration	Understand the dataset	Validated tables and relationships
Sales Analysis	Analyze revenue and orders	Identified sales trends
Product Analysis	Evaluate products	Compared product performance
Marketing Analysis	Evaluate traffic sources	Measured channel effectiveness
Website Analysis	Analyze sessions	Measured website performance

7. Power BI Development Process

• Imported CSV datasets into Power BI.
• Cleaned and validated data using Power Query.
• Created relationships between Orders, Order Items, Products, Website Sessions and Refund tables.
• Built DAX measures for key business KPIs.
• Designed three interactive dashboard pages with slicers and consistent formatting.

8. Data Model

The data model follows a relational structure where Orders acts as the central transactional table. Relationships allow revenue, orders, sessions and refund metrics to be analyzed consistently across visuals.

9. DAX Measures

Measure	Purpose	Business Value
Total Revenue	Sum of sales	Tracks overall business revenue
Total Orders	Count of orders	Measures purchasing activity
Total Sessions	Website visits	Measures website traffic
Conversion Rate	Orders ÷ Sessions	Evaluates website effectiveness

Average Order Value	Revenue ÷ Orders	Shows customer spending
Refund Amount	Sum of refunds	Monitors returns

10. Executive Dashboard

KPIs include Total Revenue, Total Orders, Total Sessions, Conversion Rate, Average Order Value and Refund Amount. Monthly trend charts help management monitor performance over time, while traffic-source visuals compare customer acquisition channels.

11. Product Performance Dashboard

Compares product revenue, units sold, average order value and revenue contribution. This page helps identify best-selling products and supports inventory and promotional decisions.

12. Marketing & Website Analytics Dashboard

Analyzes sessions, orders and revenue by traffic source together with sessions by device type. These visuals help evaluate marketing effectiveness and customer browsing behaviour.

13. Key Business Findings

- Revenue and order trends provide management with a clear view of business growth over time.
- Top-performing products were identified using revenue and units sold analysis.
- Marketing channels were compared using sessions, orders and revenue to evaluate campaign effectiveness.
- Website conversion rate highlighted how efficiently visitors became customers.
- Device analysis revealed customer browsing preferences and supported digital strategy.
- Refund tracking helped monitor customer satisfaction and potential product quality issues.

14. Business Recommendations

- Increase marketing investment in high-performing traffic sources.
- Promote top-selling products through targeted campaigns.
- Review low-performing products for pricing or promotional improvements.
- Continuously monitor conversion rate to improve website performance.
- Track refund trends regularly to identify operational issues.

15. Challenges Faced

- Creating relationships between multiple datasets.

- Developing DAX measures for business KPIs.
- Selecting appropriate visualizations for executive reporting.
- Formatting dashboards for readability and consistency.
- Handling refund calculations and data validation.

16. Skills Demonstrated

SQL, Joins, GROUP BY, Aggregate Functions, Power BI, Power Query, DAX, Data Modeling, KPI Development, Dashboard Design, Data Visualization, Business Analysis.

18. Conclusion

This project demonstrates an end-to-end Business Intelligence workflow using SQL and Power BI. It showcases skills in data analysis, KPI development, DAX, dashboard design and business insight generation. It reflects the ability to transform raw business data into actionable information for decision-making.